

Executive Summary

Business Objective

The objective of this project was to evaluate whether the business is successfully building a loyal customer base or relying heavily on continuous promotional activity for revenue generation. The analysis focused on identifying high-value customers, understanding loyalty behavior, evaluating discount dependency, and discovering patterns that influence long-term customer value.

Methodology

The project was completed using an end-to-end analytics workflow involving:

- **Python** for data cleaning, preprocessing, exploratory analysis, and feature engineering.
- **MySQL** for customer segmentation and business intelligence queries.
- **Power BI** for building an interactive executive dashboard.

Several engineered business-focused features such as **loyalty score**, **customer tier**, **high-value customer flag**, and **estimated annual customer value** were created to improve customer intelligence analysis.

Key Findings

1. Loyal Customers Exist Beyond Promotional Dependency

The analysis revealed that the business is not entirely dependent on discounts to generate revenue. Customers without discounts contributed higher overall revenue and showed comparable purchase behavior to discount users.

2. High-Value Customers Drive Significant Business Value

Approximately 25% of customers qualified as high-value customers. These customers demonstrated:

- Higher loyalty scores

- Greater repeat purchase behavior
- Higher estimated annual value

Platinum and Gold-tier customers contributed the highest revenue and customer loyalty metrics.

3. Customer Loyalty Strongly Correlates with Repeat Purchases

Customers with higher previous purchase counts consistently achieved higher loyalty scores and greater annual customer value, making repeat purchase behavior one of the strongest indicators of long-term profitability.

4. Product Category Performance Differs Significantly

The Clothing category generated the highest revenue contribution, while Outerwear showed the weakest overall performance, indicating opportunities for optimization and targeted marketing strategies.

5. Subscription Customers Show Better Engagement

Subscribed customers demonstrated significantly higher loyalty scores and estimated annual value compared to non-subscribed customers, indicating that subscription programs positively impact customer retention.

Strategic Recommendations

Based on the findings, the following business recommendations are proposed:

- Reduce dependency on blanket discount campaigns.
- Shift toward personalized and behavior-based promotions.
- Focus retention efforts on Platinum and Gold-tier customers.
- Strengthen loyalty and subscription programs.
- Prioritize acquisition of customers exhibiting high repeat purchase behavior.

Final Conclusion

The analysis indicates that the business is gradually building a genuinely loyal customer base rather than relying solely on promotional discounts. Long-term customer value is more strongly influenced by loyalty, repeat purchases, and subscription engagement than by aggressive discounting strategies.

The developed customer intelligence dashboard provides a scalable framework for future customer segmentation, retention analysis, and strategic decision-making.

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